Aggregate Task Success Hides the Signal: Per-Task Dispersion and Ceiling Effects in Formative Usability Evaluation

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***Abstract*—Formative usability studies almost always report an aggregate task-success rate. We argue that this summary is structurally lossy and demonstrate the loss on complete real data rather than by simulation. Ten participants completed eight standardised tasks across two moderated rounds, yielding eighty task attempts, Single Ease Question scores on a seven-point scale, and observer commentary, for a high-fidelity Indonesian-language home-services marketplace prototype. Aggregate results are strong: 93.75 percent task success and mean ease of 6.56 with a standard deviation of 1.05. We argue both numbers are close to uninformative. Computing dispersion per task shows four of the eight tasks saturated at a standard deviation of 0.42 or below — they can no longer distinguish a good design from a very good one. Only one task, locating the FAQ and contact channel, produced meaningful spread, and it is also the lowest-scoring. A matched pair of discovery tasks isolates a further effect: a well-labelled category was found without difficulty while a near-synonymous label drew a 0.60 point drop with triple the variance, separating a taxonomy defect from a layout defect. We further report that the two instrumented defect probes were the only tasks immune to saturation, which suggests a practical rule: every formative task set needs at least one probe that is expected to fail. We recommend reporting per-task dispersion and probe results rather than totals, and we publish the five provenance defects that constrain the conclusion.**

***Index Terms*—usability testing, Single Ease Question, ceiling effect, dispersion, formative evaluation, task success, measurement granularity, think-aloud.**

I. Introduction

Household maintenance services in Indonesia—air-conditioner cleaning, plumbing leak repair, pump servicing, septic tank suction—are dominated by micro and small enterprises whose customer acquisition still runs through spanduk (street banners), printed leaflets, and vehicle stickers. The channel has structural properties that are easy to overlook because they are invisible to the firm as well as the customer: a fixed physical area, which caps how much information can be presented; no record of prior work; and no price, which is negotiated privately and therefore unverifiable in advance.

A web interface removes the area constraint. It does not follow that the resulting interface is good, and it certainly does not follow that it is measurable. Most formative evaluations report a headline success rate, and a headline success rate is a lossy summary of a multidimensional, per-task, per-participant result.

This paper's contribution is to show, on complete real data, **how much a headline success rate can hide.** We find a 93.75% success rate coexisting with four saturated tasks that can no longer generate evidence, one task whose ratings span the entire scale, and a service taxonomy that misleads participants who are otherwise fluent. Our practical recommendation follows directly: **formative evaluation should report per-task dispersion and defect-probe results, not just totals.**

A. Research questions

- **RQ1.** With what ease do young adult users complete a standardised service-discovery and booking task sequence?
- **RQ2.** Which tasks discriminate between participants, and which are saturated?
- **RQ3.** How can a task instrument be constructed so that some tasks remain informative even when the design is strong?

B. Related work

Task-based design and evaluation of e-commerce interfaces is well established in the Indonesian literature [2], and Tianti et al. [1] demonstrate a small-business website raising promotional reach for a catering MSME. Sunarso et al. [3] provide evidence that visual content raises attraction and conversion in MSME digital marketing. We adopt the underlying premise while declining to import effect sizes measured on different stimuli.

The Single Ease Question (SEQ) is a widely adopted post-task instrument: one 7-point item administered immediately after each task attempt, whose only requirement is that the task be defined in terms the participant can recognise [4]. We adopt it for breadth—one comparable number per attempt—while noting a trade-off our own data exposes sharply (Sec. III-B): SEQ is a *mean* measure, and means conceal precisely the information a formative study needs. Reporting the mean alone would have led us to describe this prototype as uniformly easy, which Sec. III-E shows it is not.

Our argument is a special case of a general problem in measurement: a summary statistic computed over heterogeneous items can be dominated by items carrying little information while masking the items that carry all of it. Ceiling effects are the standard psychometric term for this, and they are widely discussed in survey research and adaptive testing, where item saturation is a known threat to item-response estimation. What appears to be missing from HCI practice is the routine reporting of dispersion at the *task* level, as opposed to the study level. A matched pair of tasks differing in exactly one property is a standard identification strategy [5]; we use one below to separate a taxonomy defect from a layout defect.

Heuristic inspection [6] and its Indonesian applications [7], [8] are complementary but operate on a different object: the code, not the experience. The convergence between the two methods is examined in our companion papers.

C. Scope and contributions

This is a *formative* evaluation of a single prototype build. No variable measuring conversion, revenue, bounce rate, cognitive load, or trust change was instrumented, and we make no claim about any of them. Our contributions are: a per-task dispersion analysis of a complete evaluation; a matched-pair design that converts a vague complaint into a specific diagnosis; a demonstration that defect probes are the saturation-immune component of a task instrument; and a candid account of five provenance defects.

II. Methods

A. Prototype under test

A high-fidelity React prototype, localised into Indonesian, implementing service discovery with search and category filtering, service detail pages, a four-stage checkout (summary → schedule → address → payment), a confirmation/invoice view, technician tracking, and an FAQ/contact area. Two properties were left unchanged during the study because they bear directly on our research questions: prices render in **USD** while the interface is Indonesian-language, and service descriptions include **technician-internal procedure content** intended for engineers rather than customers.

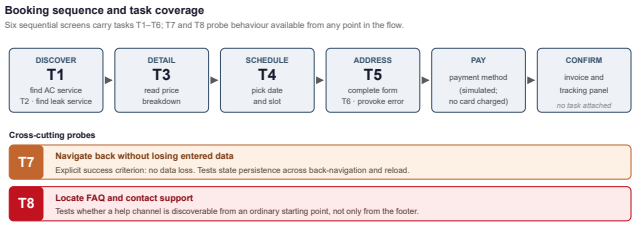


Fig. 1. The eight-task instrument mapped onto the prototype's booking sequence. Six sequential screens carry T1–T6; T7 and T8 probe behaviour available from any point in the flow.

TABLE I
Task instrument. T6 and T7 are defect probes.

Code	Task
T1	Locate the air-conditioner cleaning service
T2	Locate the leak-detection service (matched to T1)
T3	Read the detailed price breakdown
T4	Select a booking date and time slot
T5	Complete the address form
T6	Trigger and identify a validation error (**probe**)
T7	Navigate back without losing data (**probe**)
T8	Locate FAQ / contact support

B. Participants and disclosure

Ten participants were recruited and tested in two rounds of five, identified only by anonymous codes (P01–P05, P06–P10). Sessions were conducted in Semarang, Central Java, between 1 and 30 October 2026.

Disclosure on demographics. Participants were Gen Z, aged **19–21**, verified against their national ID cards by the research team at recruitment and confirmed for all ten. No date of birth was transcribed into the research data, as a privacy measure; the age range is recorded here as an aggregate. Gender and occupation were not collected. The sample is therefore described as *Gen Z users recruited purposively* and we make **no representativeness claim** beyond that description.

C. Tasks

Eight tasks spanning the booking flow (Fig. 1). T1 and T2 are a *matched* discovery pair differing only in service category. **T6 and T7 are defect probes**—tasks with explicit failure criteria, designed so that a correct prototype yields failures rather than successes.

D. Measures and procedure

Task outcome: completed / completed with assistance / not completed, against a written criterion per task. *SEQ:* 1 = *very difficult* … 7 = *very easy*, immediately after each attempt. *Observer commentary:* notes on confusion, verbalised expectations, and unsolicited suggestions.

Sessions were moderated with concurrent think-aloud (Fig. 2). Screen and audio recording was conducted with the consent of all ten participants and is stored under institutional control, scheduled for deletion on 1 January 2027; **the**



Fig. 2. A testing session in progress. A participant (seated, left) works through the task sequence while the moderator (foreground, right) guides the session. Additional participants and moderators are visible in the background, reflecting the two-round testing schedule. Faces blurred for privacy.

TABLE II
TASK SUCCESS, BOTH CODINGS.

	Round 1	Round 2	Combined
As recorded	39/40 (97.5%)	40/40 (100 %)	79/80 (98.75 %)
Corrected (head)	37/40 (92.5 %)	38/40 (95.0 %)	75/80 (**93.75 %**)

recordings were not used in this analysis, and all qualitative material available to it is therefore observer paraphrase, not verbatim transcript. We use quotation marks in this paper only for source documents, never for participant speech.

E. The completion-rate dispute, and how we resolved it

Completion was coded during sessions and recorded as 39/40 (round 1) and 40/40 (round 2). Both source recap sheets then appended an auditor note that specific attempts should have been coded failures: round 1, two T6 attempts recorded as successes although the criterion was *find the validation message* and the notes state the participants did not find it; round 2, one T6 attempt recorded as a success although the note reads "user did not find any error," and another reaching confirmation with a street address consisting only of the characters (and).

This is a genuine interpretive disagreement about whether failure on a *sub-criterion* constitutes task failure, not an arithmetic error. Rather than adopt either position silently, **we report the auditor-corrected figures as the headline and present the as-recorded figures as a sensitivity analysis** (Sec. III-A).

III. RESULTS

A. Aggregate task success

Table II. All five failures occurred at T6 or T8. A rate above 90% would ordinarily indicate a largely successful design. **We do not read it that way**, for two reasons. First, the T6 success criterion was permissive—a permissive criterion inflates the aggregate, so the headline number is an upper bound on its own reliability. Second, and decisively, success is *ceiling-bound*: four tasks were near-universally completed, so the metric cannot discriminate between design states.

TABLE III
COMPLETE SEQ MATRIX ($n = 10$ PARTICIPANTS $\times$ 8 TASKS).

	T1	T2	T3	T4	T5	T6	T7	T8	Mean
P01	6	5	7	7	7	7	4	6	6.13
P02	7	7	7	7	7	7	7	7	7.00
P03	7	5	7	7	7	7	7	3	6.25
P04	7	7	7	7	7	7	7	7	7.00
P05	7	5	7	7	7	3	7	1	5.50
P06	6	6	6	6	6	7	7	5	6.13
P07	7	6	7	7	7	7	7	7	6.88
P08	7	7	7	7	7	7	7	7	7.00
P09	7	7	7	7	7	7	7	7	7.00
P10	7	7	6	7	7	6	7	7	6.75
M	**6.80**	**6.20**	**6.80**	**6.90**	**6.90**	**6.50**	**6.70**	**5.70**	**6.56**
SD	0.42	0.92	0.42	0.32	0.32	1.27	0.95	2.11	1.05

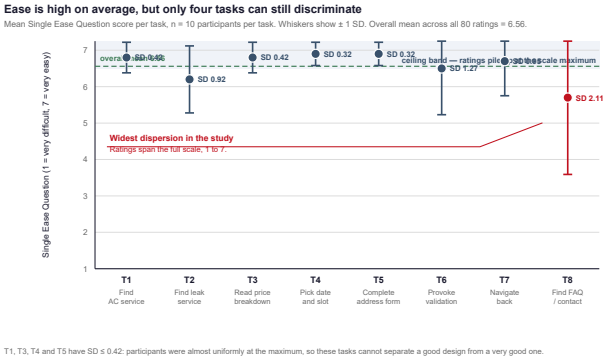


Fig. 3. Mean SEQ per task with ± 1 SD whiskers. Four tasks sit against the scale ceiling with $SD \leq 0.42$. T8 has both the lowest mean and the widest dispersion, spanning the full 1–7 range.

B. Ease, and why the mean is not the finding

Overall SEQ: $M = 6.56$, $SD = 1.05$ ($n = 80$ task-level ratings), median 7. Distribution: 62 ratings of 7, 10 of 6, 4 of 5, 1 of 4, 2 of 3, 1 of 1. Per-task means and dispersions are given in Fig. 3 and Table III.

Two observations follow, and **neither is visible in the aggregate mean.**

(a) Four of eight tasks are saturated. T1, T3, T4, and T5 have $SD \leq 0.42$ on a 7-point scale; for T4 and T5 no participant rated below 6. These tasks cannot separate a good design from a very good one, and further iterations must not rely on them for evidence.

(b) T8 is the study's real signal. Support-channel discoverability has the lowest mean ($M = 5.70$) *and* by far the widest dispersion ($SD = 2.11$), spanning the full usable range from 1 to 7. Individual comments locate the cause precisely: participants found the WhatsApp control and read it as a *booking* action rather than a *support* channel; some located help only in the footer; one found no direct contact number at all. Two participants rated T8 at 7; one rated it 1.

A task whose ratings span the entire scale is underserved by the design, not merely unfamiliar to the participant. A wide spread invites the reading "some users need

TABLE IV
DEFECT PROBES VERSUS ORDINARY TASKS.

Task	Type	M	SD	Range
T6	probe	6.50	**1.27**	3–7
T7	probe	6.70	**0.95**	4–7
T1,T3,T4,T5	ordinary	—	0.32–0.42	6–7

training." The comments refute it: the mechanism is not comprehension but *labelling and placement*—a control that is present, visible, and means something other than what it appears to mean.

C. How many saturated tasks make an instrument uninformative?

The practical question behind (a) is a counting one, and this study's proportions let us answer it descriptively. **Four of eight tasks—half the instrument—produced** $SD \leq 0.42$**.** Those four contributed 32 of 80 observations, and their within-task variance is negligible, so they contribute almost nothing to the study-level SD of 1.05. The entire reported dispersion is generated by four tasks: T2, T6, T7, and T8.

This has a direct consequence for instrument design. If a formative study wants per-task dispersion to carry diagnostic information, then **at least half its tasks must remain discriminative**, or the study-level variance statistic will be reported as though it described the whole instrument when it in fact describes a minority of it.

We note this is an observation from one study, not a general threshold. The proportion of saturated tasks depends on instrument difficulty, participant expertise, and design quality. What generalises is the reporting requirement: **the proportion of saturated tasks should be stated, because it determines how much of the study-level variance is informative.**

D. Defect probes are the saturation-immune component

The two tasks designed with explicit failure criteria behaved as intended, and this is the study's most transferable methodological finding (Table IV).

Both probes sit far above the four saturated tasks in dispersion, and T6 carries the third-lowest mean in the instrument. **A probe did not saturate even though the design is strong.** The reason is structural rather than accidental: an ordinary task asks whether the user can *succeed*, and a competent prototype lets nearly everyone succeed, which necessarily compresses variance. A probe asks whether the user can *fail in a specific way*, which a competent prototype permits precisely because the defect—or the absence of one—is detectable.

We therefore propose a practical rule: **every formative task set should include at least one defect probe with an explicit failure criterion.** Without one, a study of a strong design has no instrument that can continue generating evidence, and its remaining tasks will saturate.

E. The matched pair isolates a taxonomy defect

T1 and T2 are the same task with a different category. T1 (find AC cleaning) reached $M = 6.80$**,** $SD = 0.42$. T2 (find leak detection) reached $M = 6.20$**,** $SD = 0.92$—a 0.60-point drop, with triple the variance, on an otherwise identical task.

Participants attributed the difficulty to near-synonymous service labels ("smart leak detection" versus pipe inspection) that the interface did not differentiate, and in two cases to the booking control sitting below the fold. This is a **labelling** problem, not a **layout** problem: the category was reachable, but participants could not determine *which* entry addressed their problem. The practical consequence is disproportionate—merging two near-synonymous catalogue entries is a data change, while redesigning the discovery screen is a redesign.

F. A contradiction between our own sources

The narrative report's recommendation matrix rates support/contact remediation **Low** priority. Both recap sheets rate it **High**. The narrative's own session data contradicts its own prioritisation, since four of its sessions record difficulty locating support. We adopt the recap sheets' rating, on the grounds that it is supported by the largest weight of recorded evidence and by the dispersion analysis above. We report the disagreement rather than suppressing it, because a reader checking our recommendation against the primary documents should be able to see that we noticed.

IV. DISCUSSION

A. What worked

The prototype's *structure* is sound. Four tasks sit at or near ceiling with almost no dispersion. The booking flow follows an expected mental model—browse, detail, schedule, address, pay, confirm—and participants navigated it without instruction.

Note the asymmetry in T5: the address form reached ceiling ($M = 6.90$, $SD = 0.32$) **while generating the most frequently requested features in the entire study**—geocoding, autocomplete, saved addresses. The form is easy to fill and wrong in design. **Ease of completion and fitness for purpose are independent properties, and SEQ measures only the first.**

B. Methodological recommendations

- **Report per-task dispersion, not just aggregate success.** A 93.75% success rate coexisted here with a task rated as low as 1/7. The dispersion analysis is what surfaced it.
- **State the proportion of saturated tasks.** It determines how much of the study-level variance is informative (Sec. III-C).
- **Include at least one defect probe.** Ordinary tasks saturate on a competent design; probes do not (Sec. III-D).
- **Prefer matched task pairs to isolate causes.** The T1/T2 contrast converted a vague complaint into a specific, cheap-to-fix diagnosis.
- **Treat effortful completion as a smell.** The address form was easy to fill and wrong. Comfort is not fitness.

C. Threats to validity

T-1 [Severe] There is no consensus on participant list between the source documents. Ten participants were recorded in the narrative corpus; a somewhat different group of ten participants were reported in the two quantitative recap documents. *Consequence:* it cannot be proved that the SEQ and the narrative corpus are about the same ten people. They are never combined together.

T-2 [Severe] Lack of identification information on prototype build, date, and moderator. It was required to record the prototype and its build hash before the start of any sessions. *Consequence:* no result can be linked to the current prototype.

T-3 [Low] Demographic information is not detailed enough. The participants were Generation Z, ages 19–21, with their identity verified by national ID cards during recruitment; no date of birth was recorded and gender/occupation information was not gathered. *Consequence:* the makeup of the sample is known up to that.

T-4 [Severe] Ethics and consent. The written informed consent was obtained from all ten participants and safeguard procedures used in each session are known. **No Institutional Review Board approval was sought beforehand**, which diverges from our protocol and BINUS requirement for research with human participants to be approved before recruitment. A determination has been requested from the Research Ethics Committee under the exemption track for secondary analysis of deidentified data; the request and a full disclosure of the departure are documented in ethics-request/. **Until a written determination is issued, this threat remains open.** Consent and institutional review are separate requirements, and satisfying the first does not discharge the second.

T-5 [Moderate] Ceiling effect limits four tasks. T1, T3, T4, T5 cannot discriminate between design states.

T-6 [Moderate] Variance collapse between rounds. Round-2 per-task SD (0.49) is $2.8\times$ smaller than round-1 (1.39). Possible causes cannot be distinguished from the available records, and the build did not change between rounds. We draw **no** design-effect conclusion.

T-7 [Moderate] Qualitative corpus is observer paraphrase, not transcript. *Consequence:* no direct quotation is used anywhere.

V. Conclusion

We evaluated an Indonesian-language home-services marketplace prototype with ten participants across eight tasks and two rounds. Task success was 93.75% and mean ease 6.56/7—and the most useful result of the study is that **neither number characterises the design.**

Beneath the aggregate sit four saturated tasks that can no longer generate evidence, one task whose ratings span the entire scale because a help channel is present but mislabelled, and a matched pair of discovery tasks that isolates a taxonomy defect from a layout defect. The two defect probes were the only tasks that stayed informative on a competent design, which is why every formative task set should carry at least one.

The methodological lesson generalises beyond this prototype. **A formative study that reports only aggregate success rates will systematically under-report exactly the problems that matter most**, because the tasks that discriminate are outnumbered by the tasks that have saturated. We recommend publishing per-task dispersion as a matter of course, and we have published ours—including the five provenance defects that constrain them.

Dataset Sharing Disclosure

The ten participants and eighty task attempts reported here are the same dataset used in two companion papers: one on price-integrity auditing and its reproduction, and one on a friction typology and remediation matrix for MSME service marketplaces. All three draw on one prototype build. The split is thematic, not evidential: **no number is unique to this paper**. Unique to this paper are the aggregate-versus-dispersion argument, the saturated-task proportion, the saturation-immunity of defect probes, and the complete rating matrix (Table III).

Author Contributions

Meutia Braniwati: Conceptualization, Project administration, Methodology, Writing – review & editing. Bagaskoro Saputro: Methodology, Investigation, Formal analysis, Writing – original draft. Aditya Paramananda: Software, Methodology, Resources, Writing – review & editing. Freysia Chandra Saliman: Investigation, Data curation, Resources, Writing – review & editing. Stephanie Evangelina Pramudito: Investigation, Project administration, Resources, Visual design, Writing – review & editing. Tan, Michael Wilson Setiawan: Software, Resources, Writing – review & editing. All authors read and approved the final manuscript.

Data Availability

All data underlying every figure will be deposited on OSF at https://doi.org/10.17605/OSF.IO/45YZJ, under the pseudonymous participant codes P01–P10 used here. Screen and audio recordings are not shared: they were not used in this analysis and are deleted on 1 January 2027 under the approved consent instrument.

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APPENDIX

"After completing the task, how easy or difficult was that task for you?" $1 =$ very difficult $\cdots$ $7 =$ very easy. Administered immediately after each of T1–T8. Ordinal scale; mean and SD reported for comparability with common practice, with the caveat in Sec. IV-C(T-5).

The complete SEQ matrix appears in the body of the paper (Table III). Participants anonymised per Sec. II-B. Round 1 = P01–P05; Round 2 = P06–P10. Values reproduced exactly as recorded in the source sheets; recomputation confirmed every reported mean. See ut-results-integration.md for the full provenance record.

PROHIBITED FIGURES

The superseded draft contained 87.5%, 4.20/5, 84.6%, and 4.23/5. These were never derivable from any source and remain permanently barred. Note that 84.6% is **not** the corrected completion rate (93.75%) and 87.5% is **not** any figure in Sec. III-A; the resemblance is coincidental and must not be used to rehabilitate them.